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LNSM

# Project 2025

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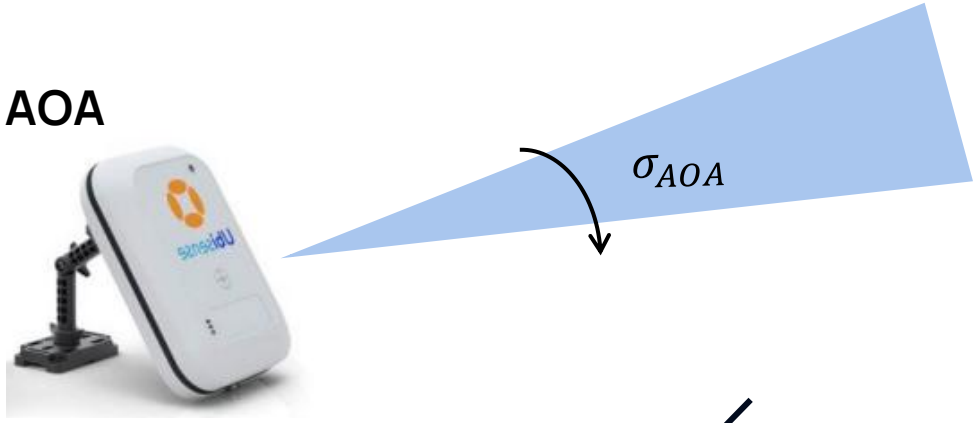
# Data visualization

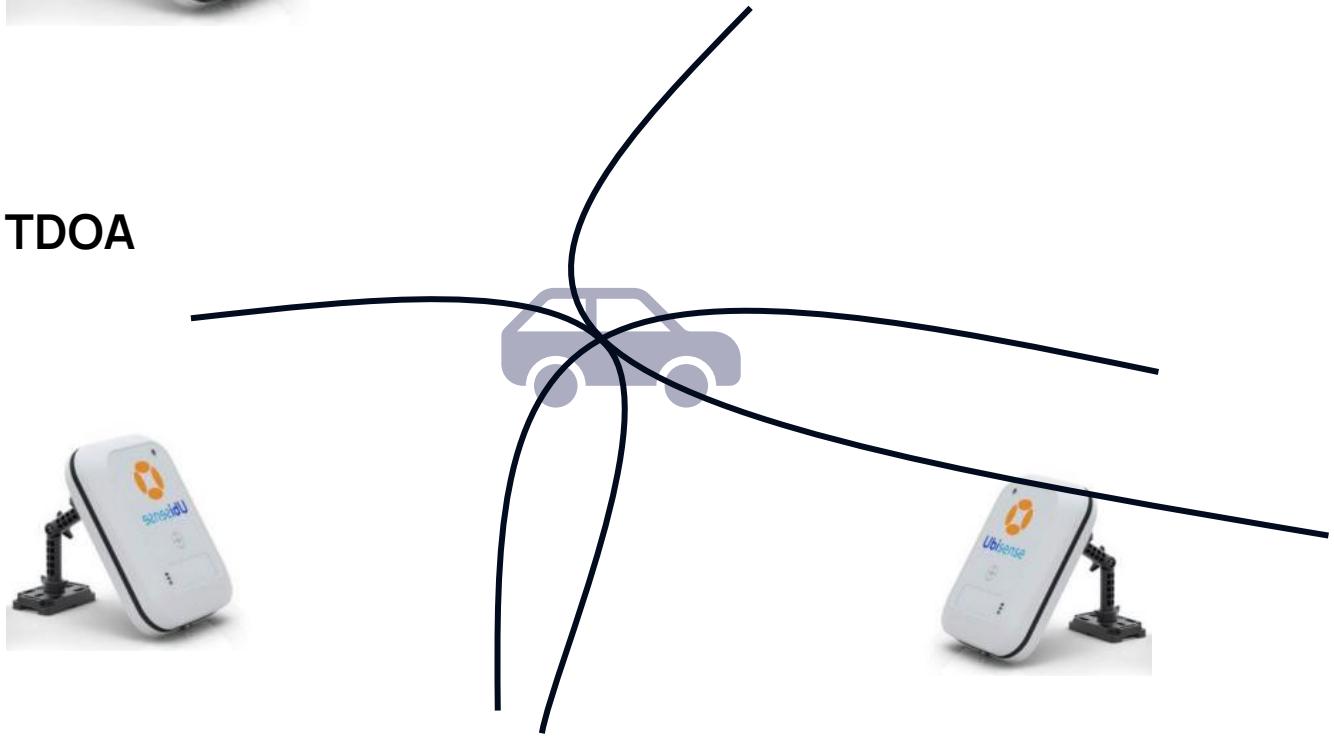
01

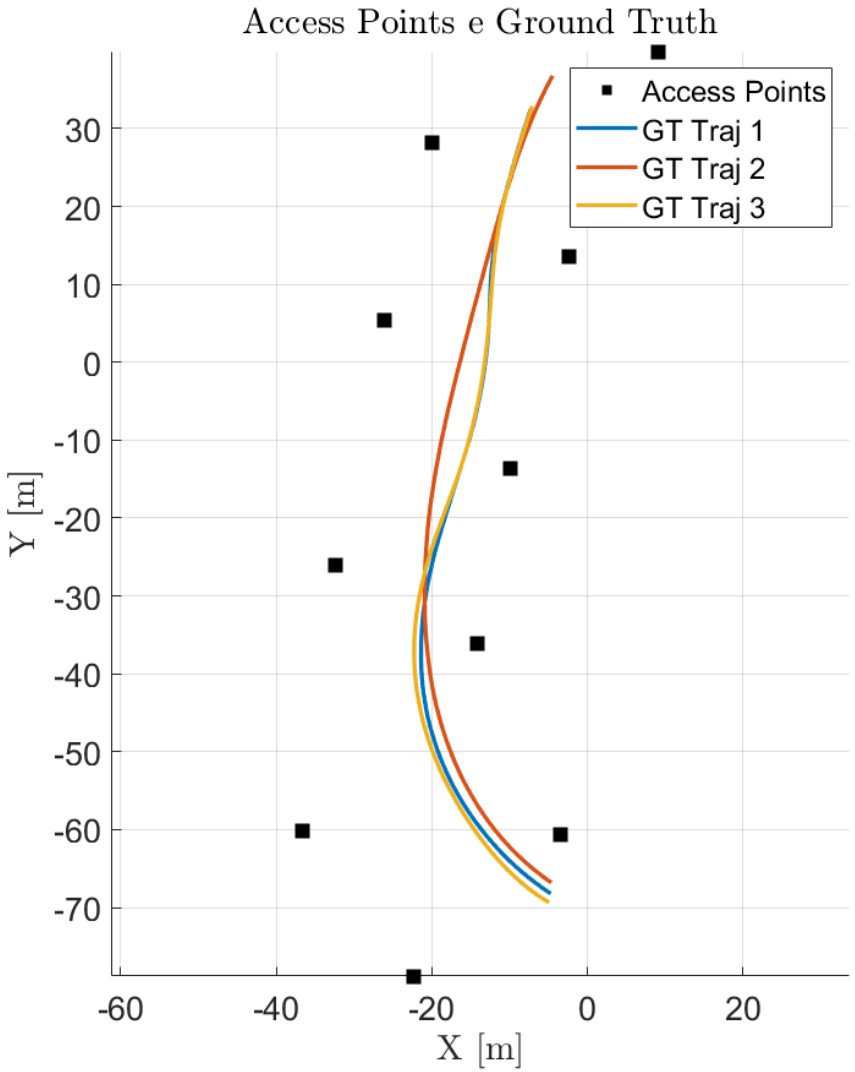
# Ground Truth – Experiment setup

**AOA**



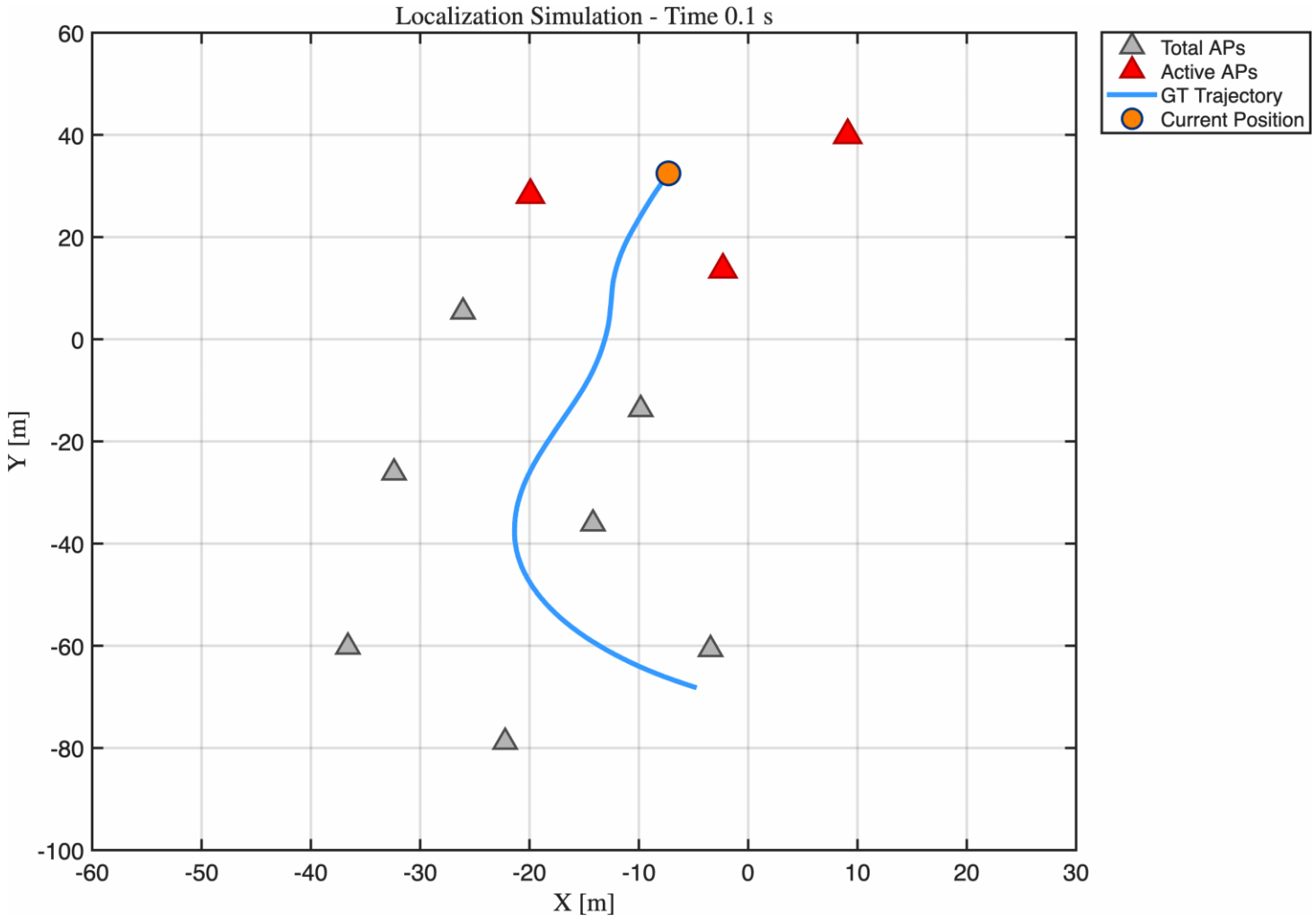

**TDOA**





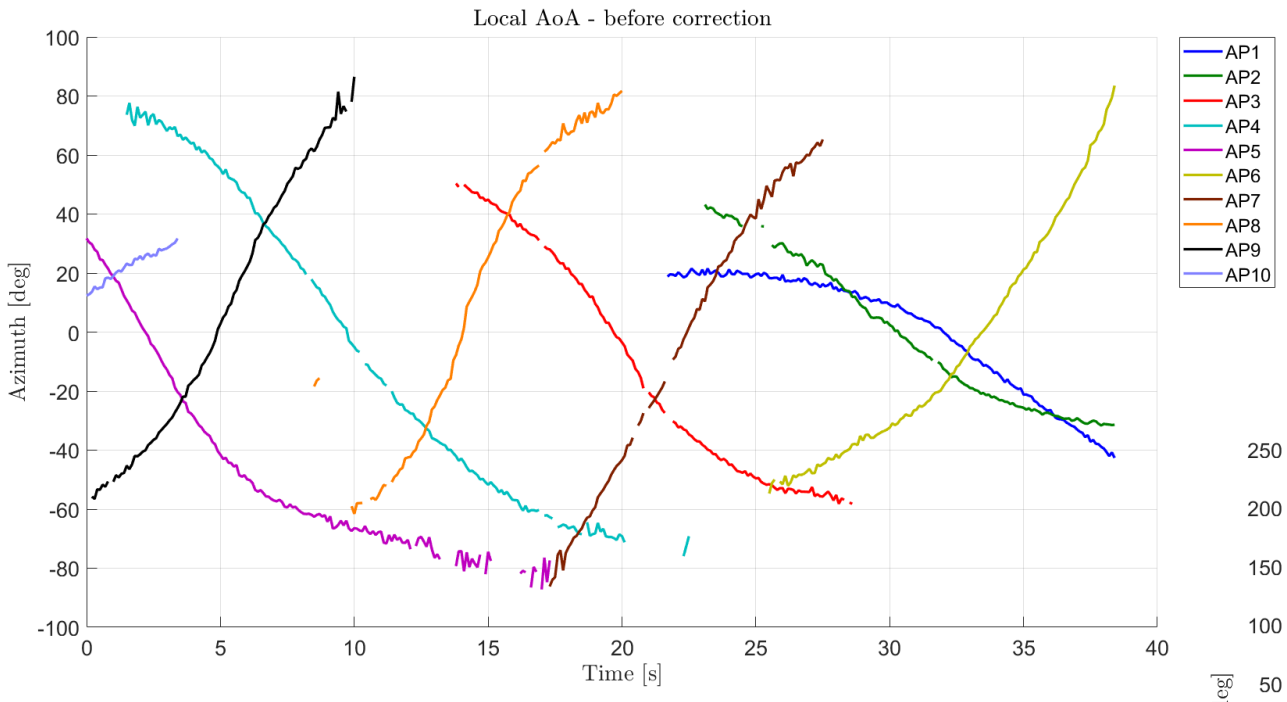
# How to deal with APs availability?

- Not all APs active at the same time;
- At each position only non-NaN measurements are used;
- Active APs depend on measurements type (AOA/TDOA);
- Accuracy changes with the number of active APs.



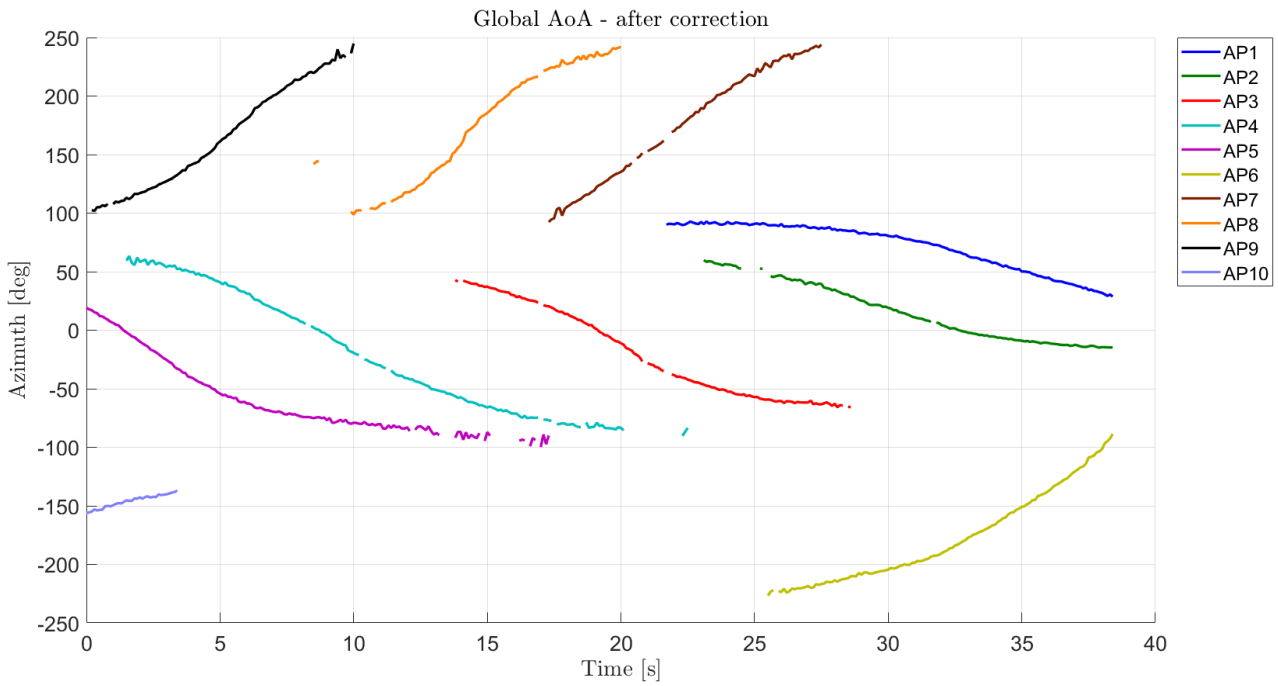

*Animation example for TDOA measurements*

# AOA data correction




- AOA raw measurements are local with respect to the antenna reference system.
- To get the global measurements we add the yaw corrections.

$$AOA_{global} = AP_{local} + AP_{yaw}$$



# 2D Localization

02

# Simulation setup

## Methodology

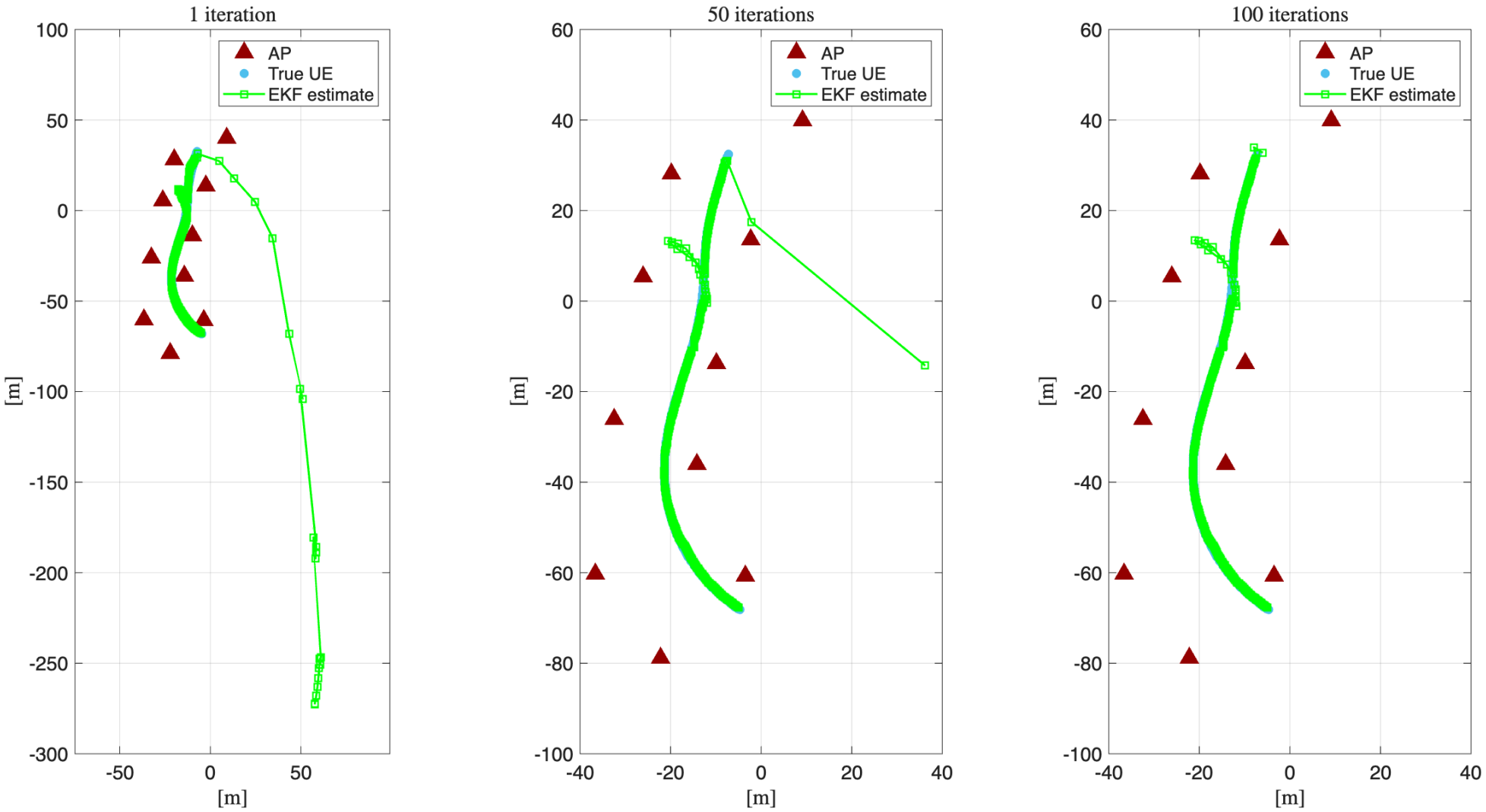
- Extended Kalman Filter (EKF) for state estimation;
- Nearly Constant Velocity (NCV) motion model.

## Parametrization

- $\sigma_{TDOA}$: $3\ \mathrm{m}$;
- $\sigma_{AOA}$: $3°/10°$;
- $\sigma_{driving}$: $2\ \mathrm{m}/s^2$.

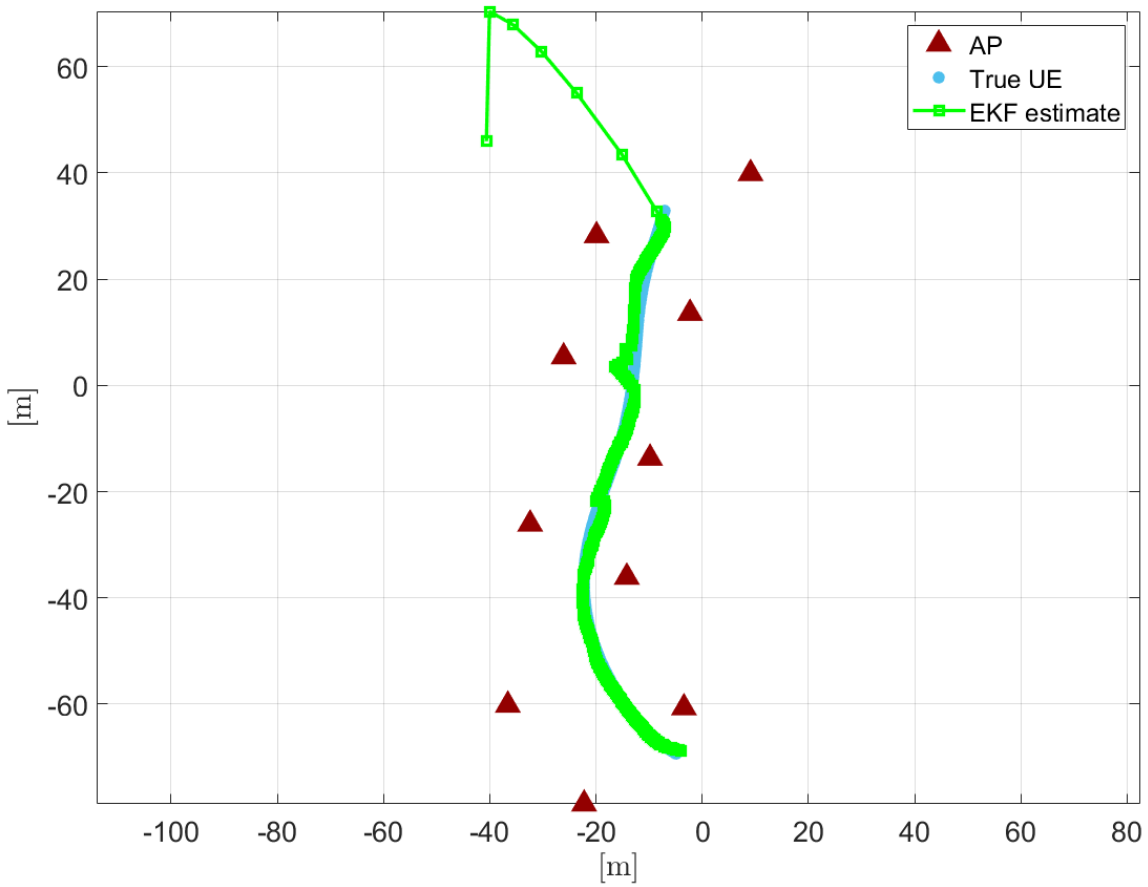
# Localization: TDOA

Trajectory estimation: $\sigma_{TDOA} = 3\ m$

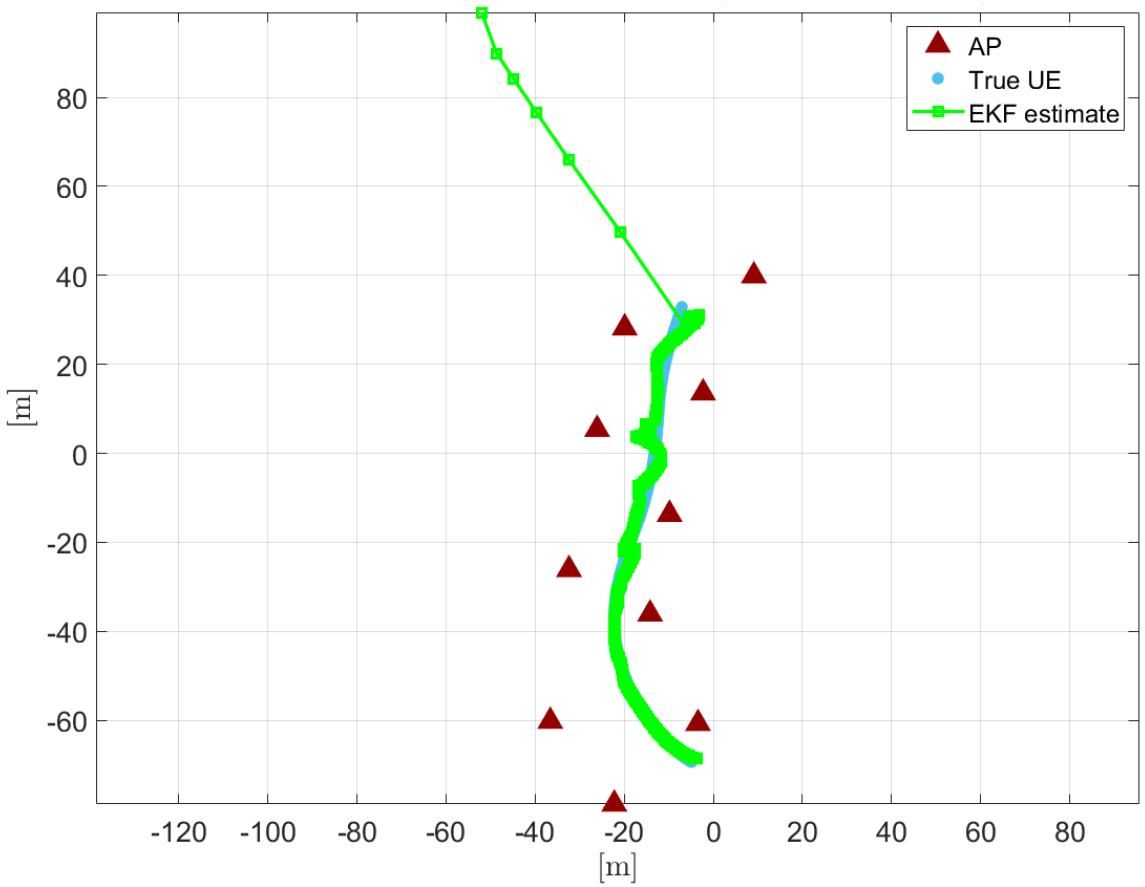


# Localization: AOA

Trajectory estimation: $\sigma_{AOA} = 10°$

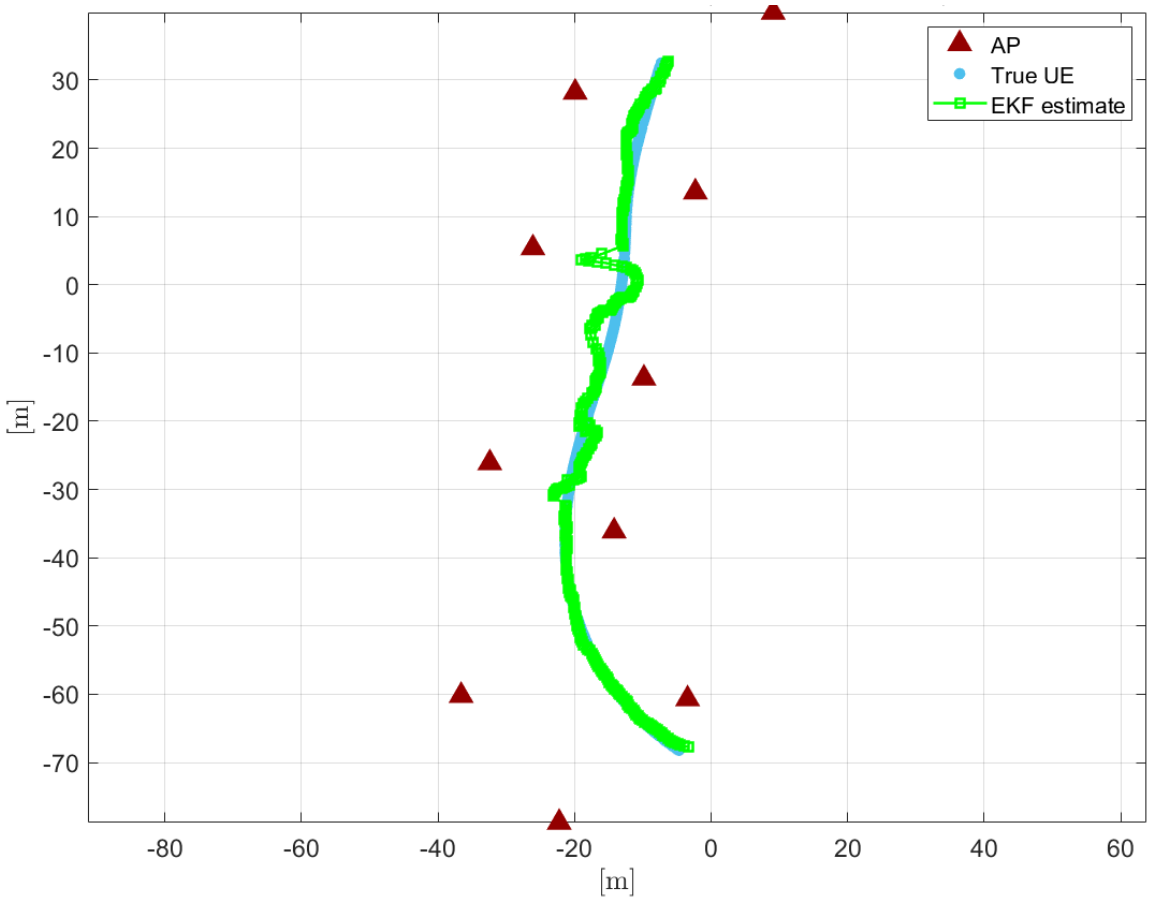



Trajectory estimation: $\sigma_{AOA} = 3°$

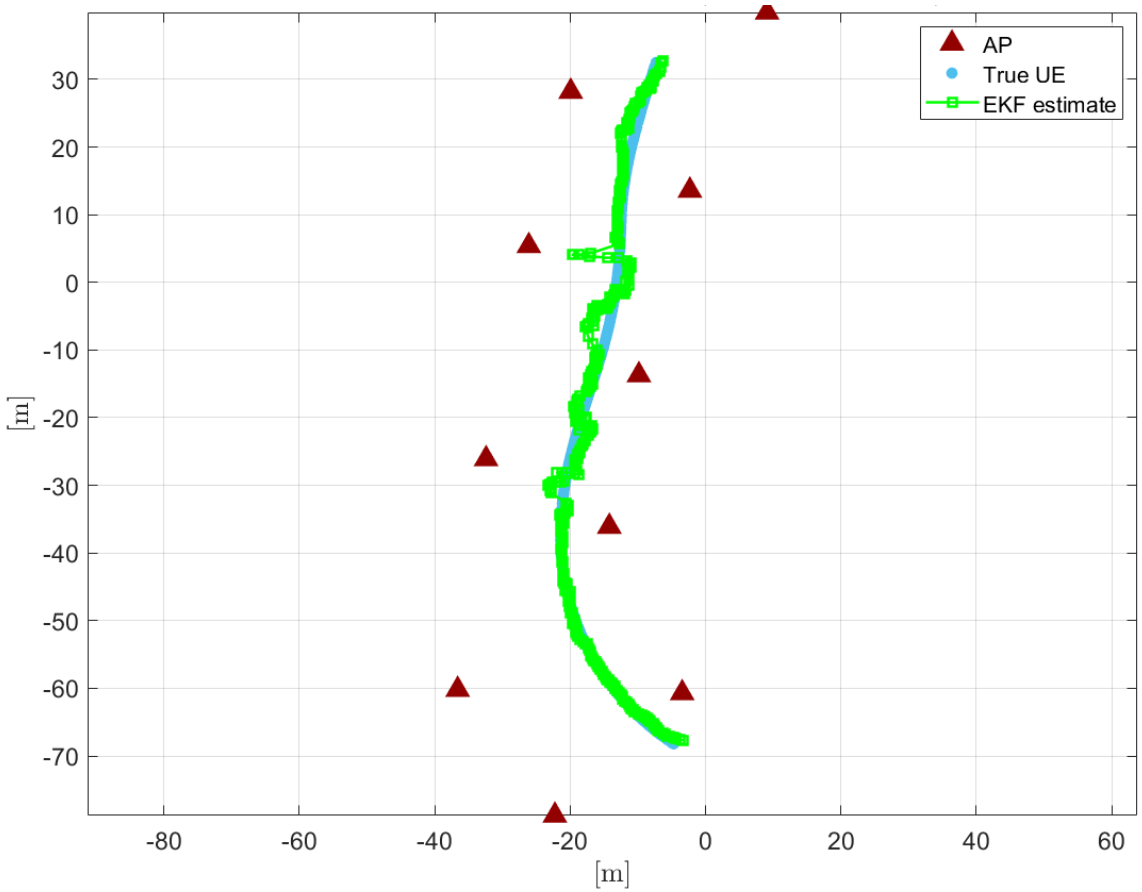


# Localization: AOA – 100 NLS iterations

Trajectory estimation: $\sigma_{AOA} = 10°$

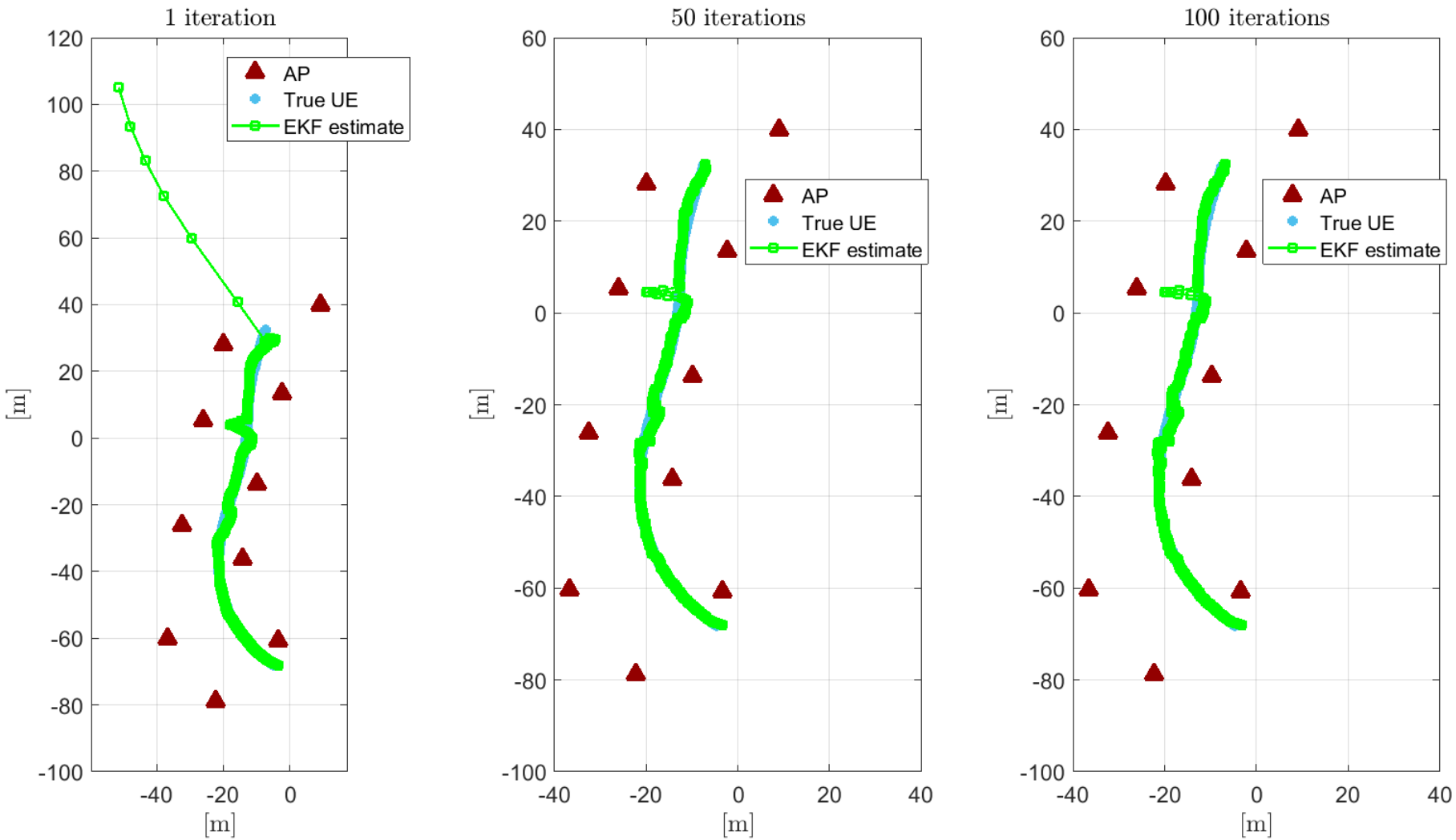



Trajectory estimation: $\sigma_{AOA} = 3°$



# Localization: TDOA + AOA

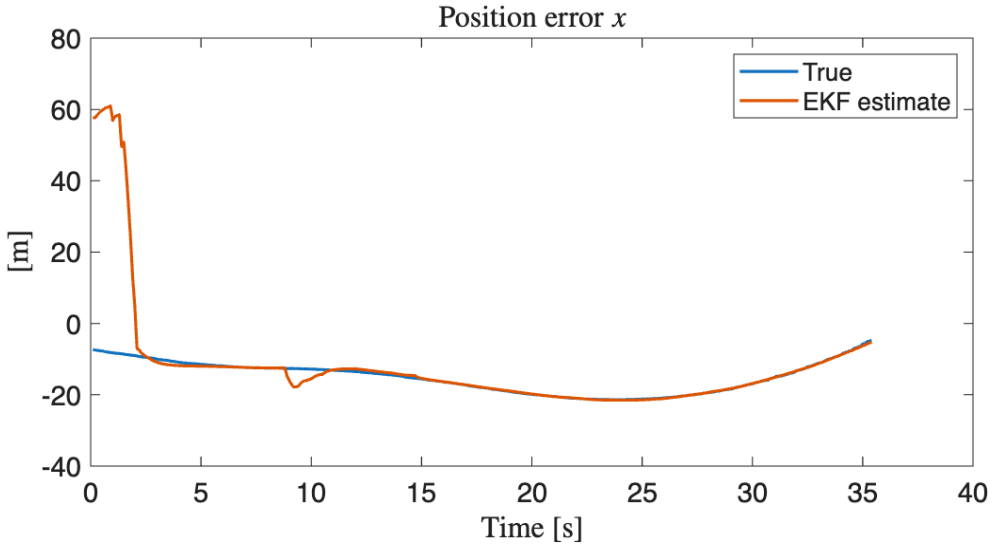
Trajectory estimation: $\sigma_{TDOA} = 3\ m; \sigma_{AOA} = 3°$

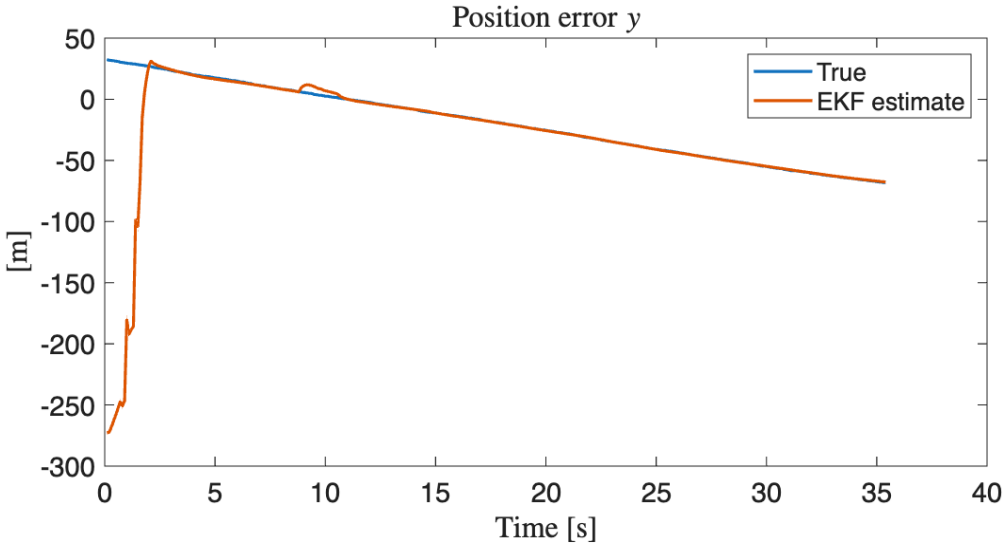


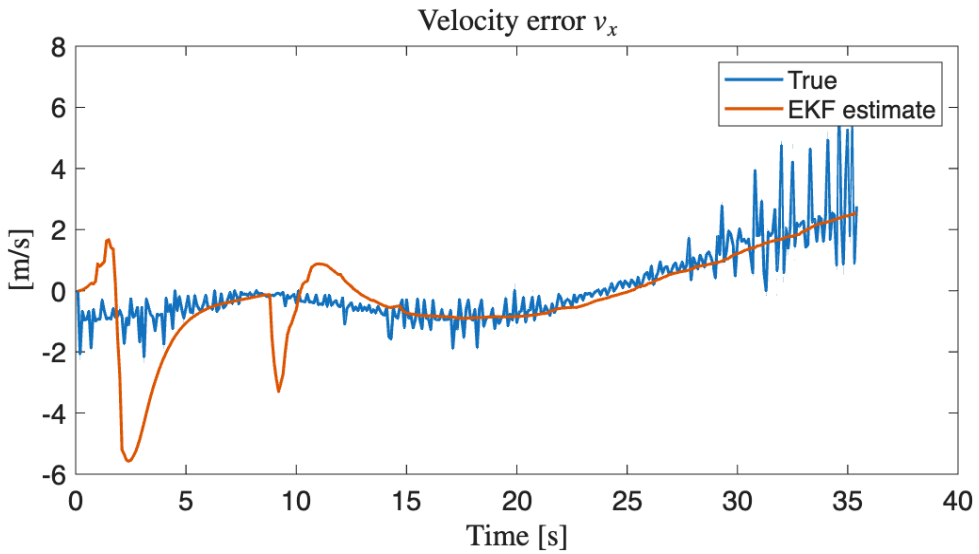
# Accuracy evaluation

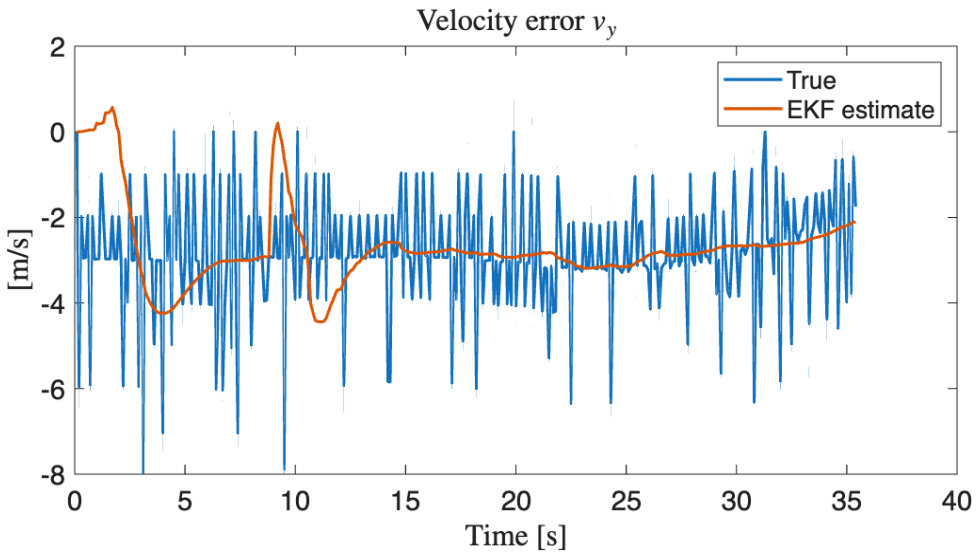
03

# TDOA: 1 iteration

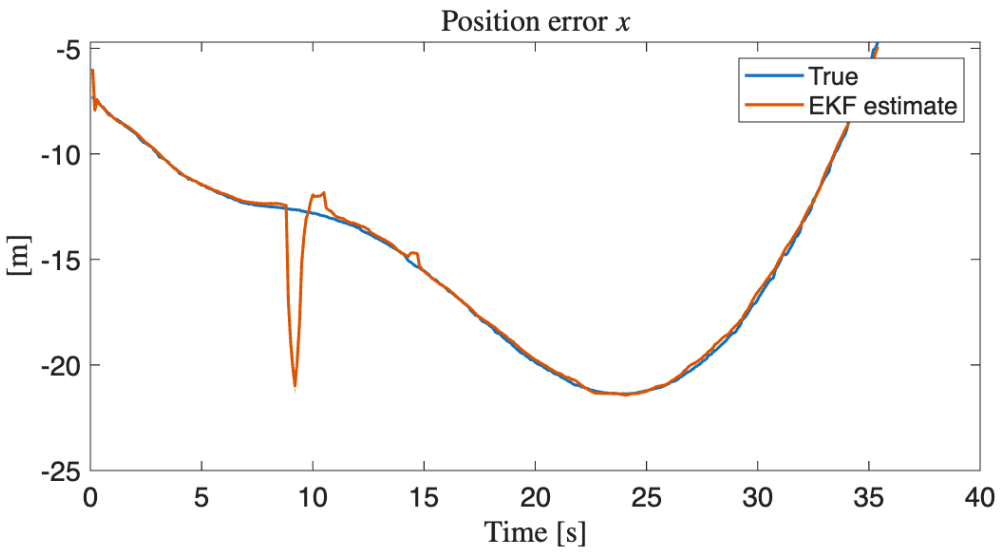


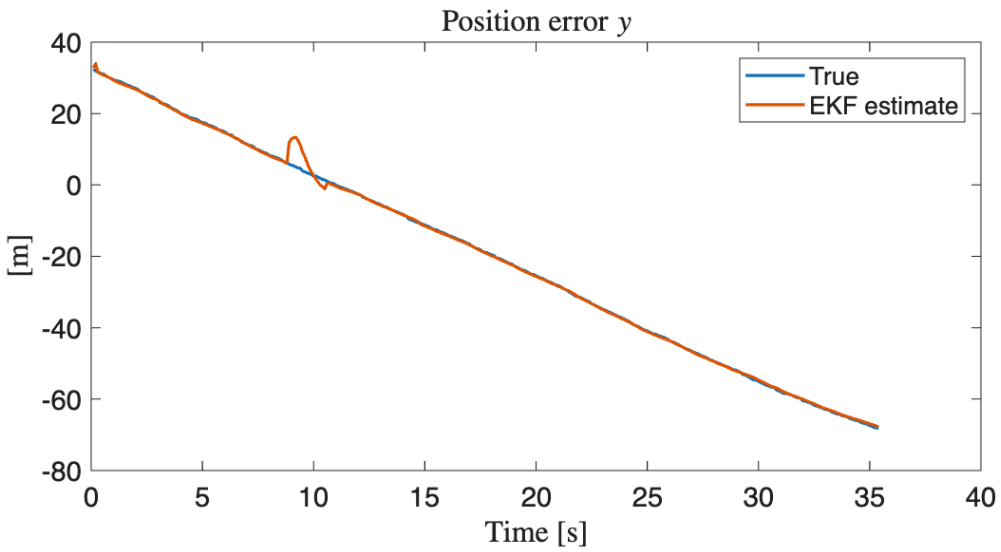



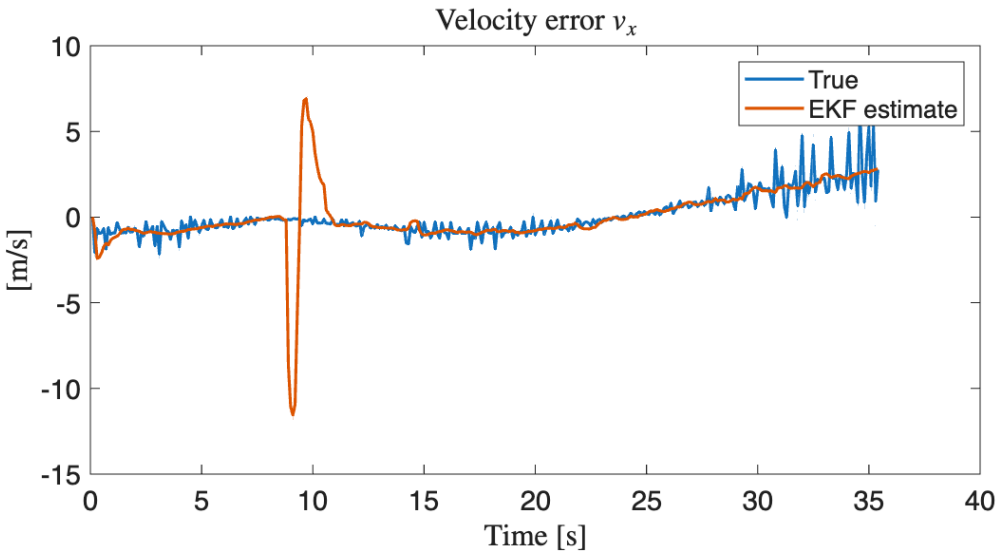



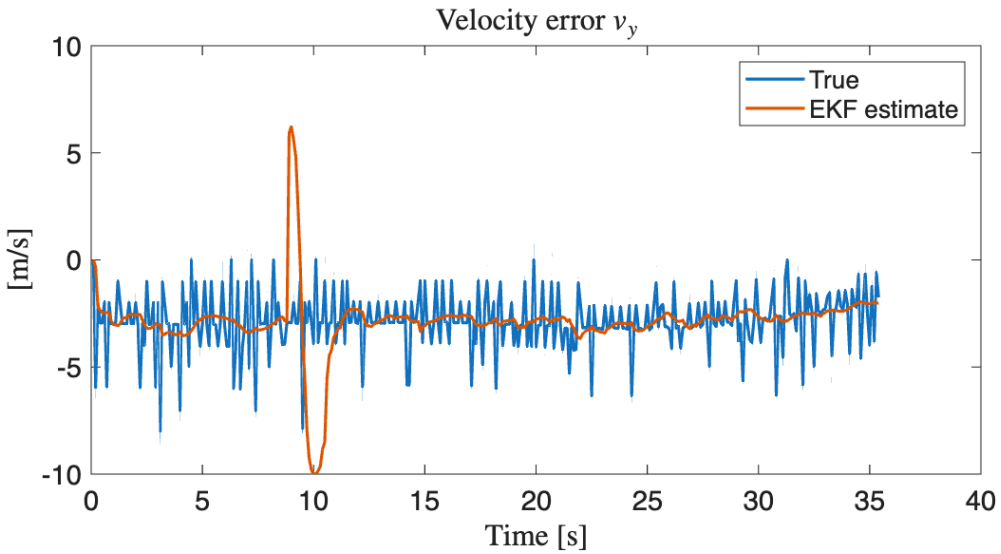



# TDOA: 100 NLS iterations

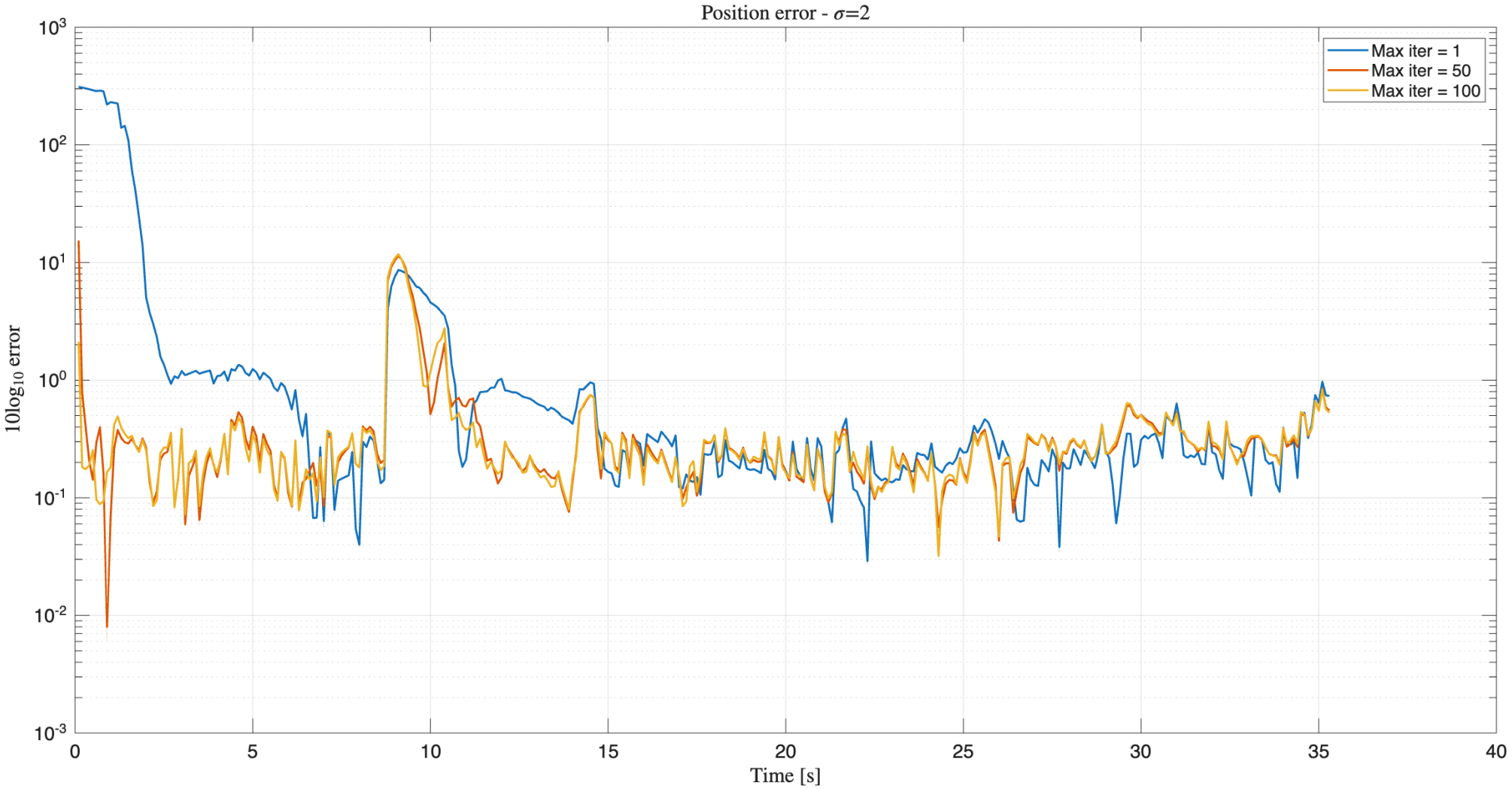


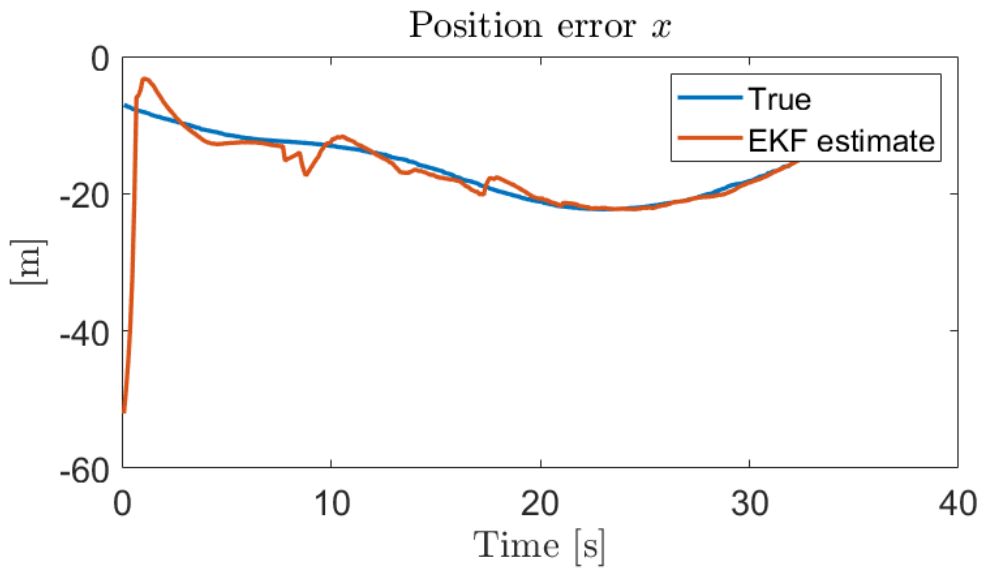



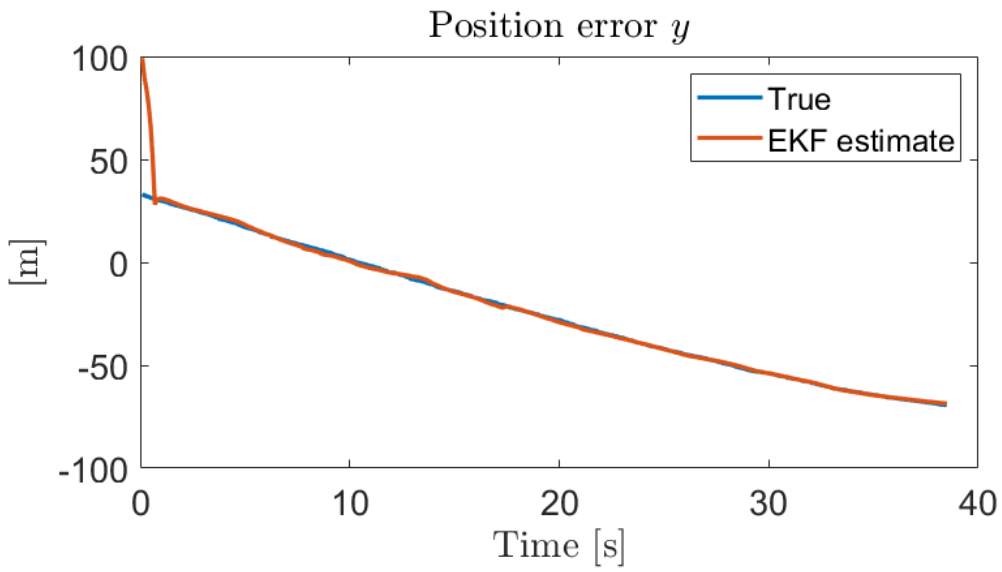



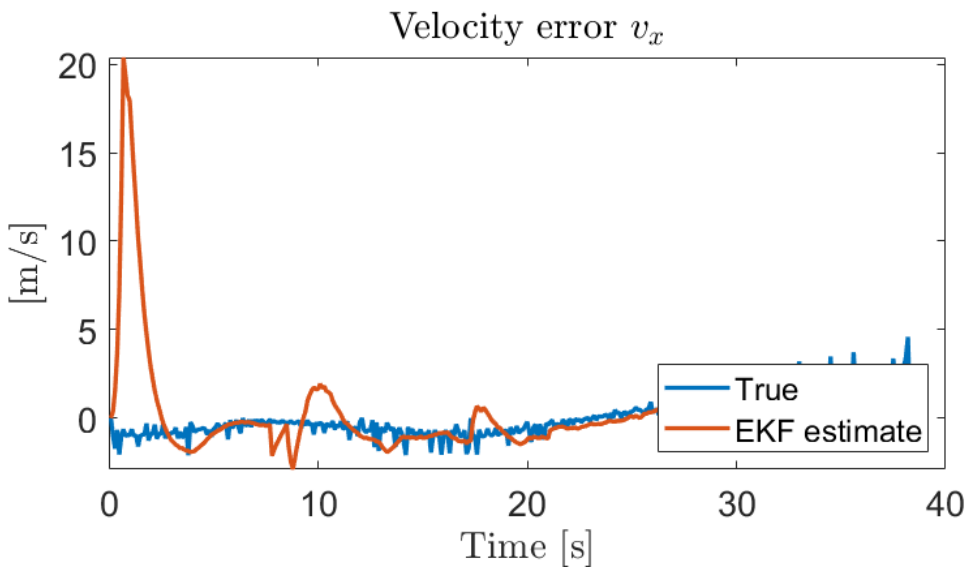
# TDOA: position error vs iterations

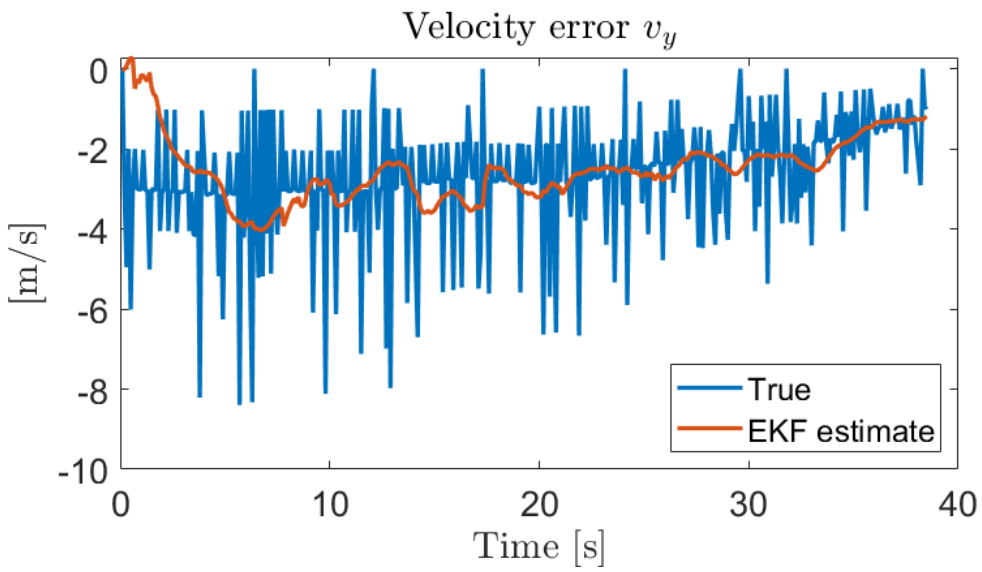


# AOA: 1 iteration

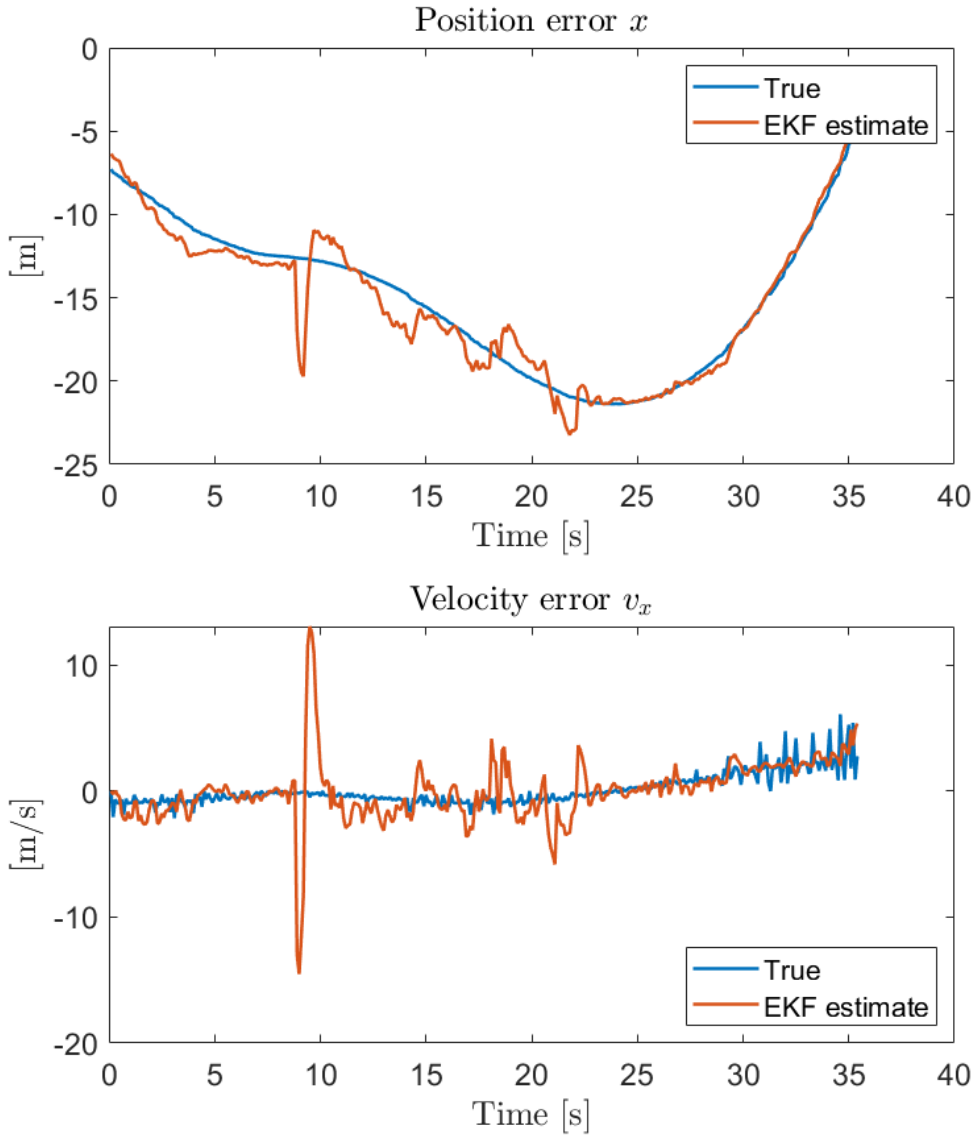


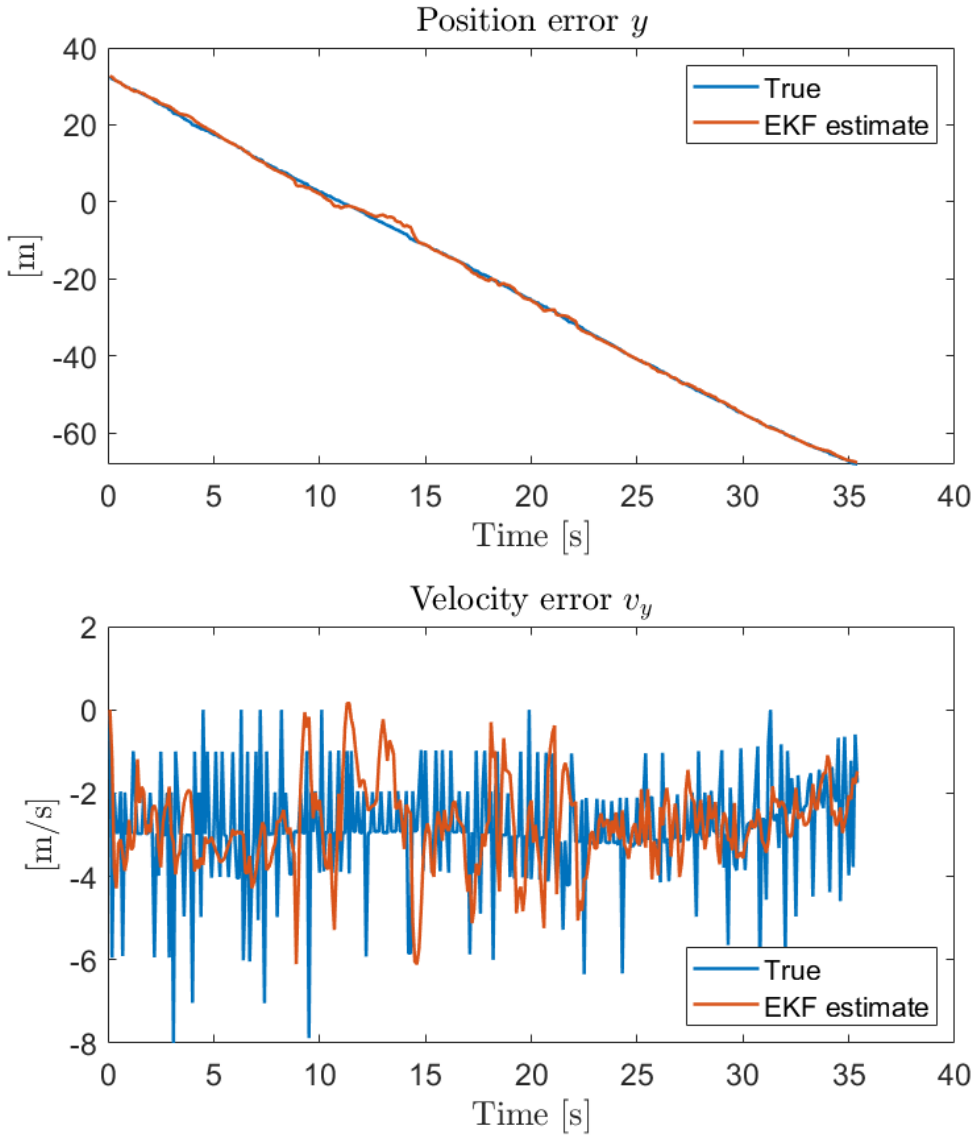



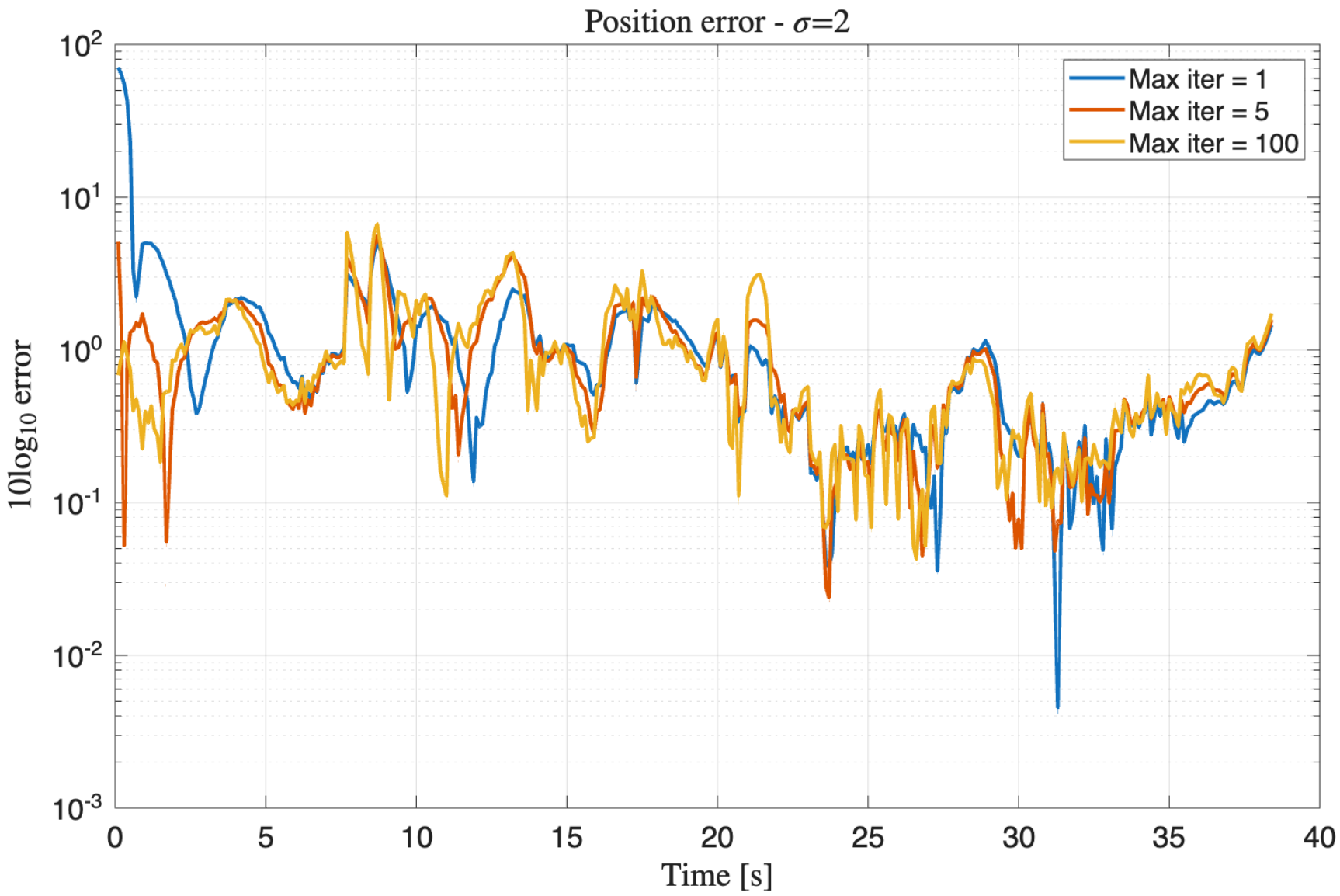



# AOA: 100 NLS iterations

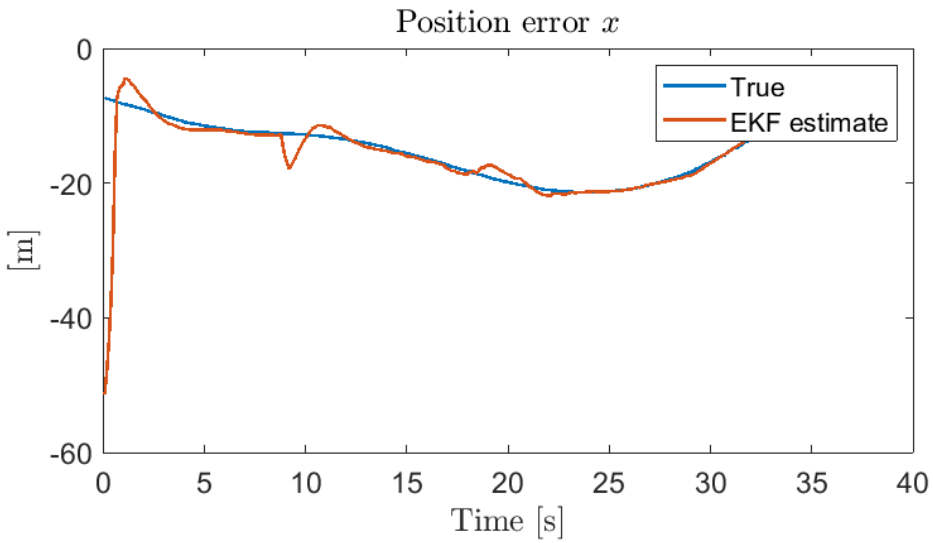


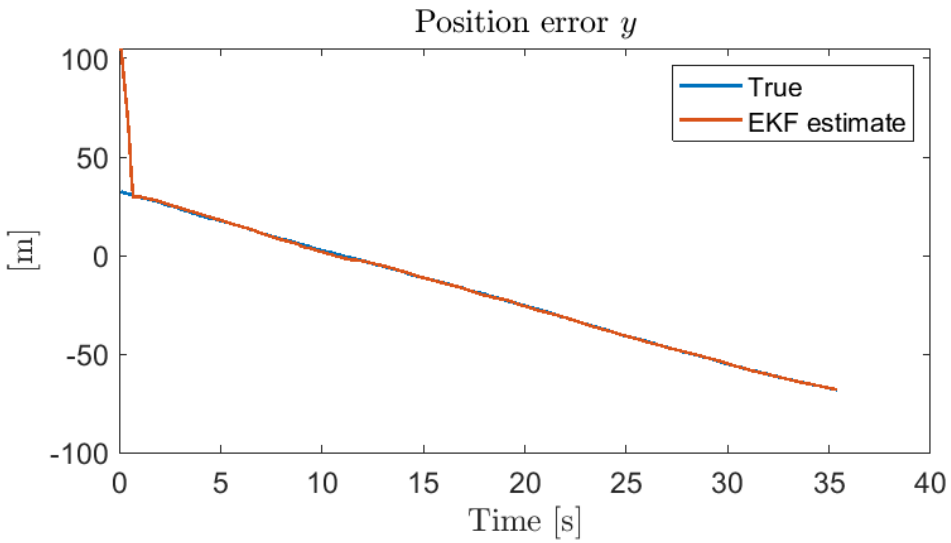



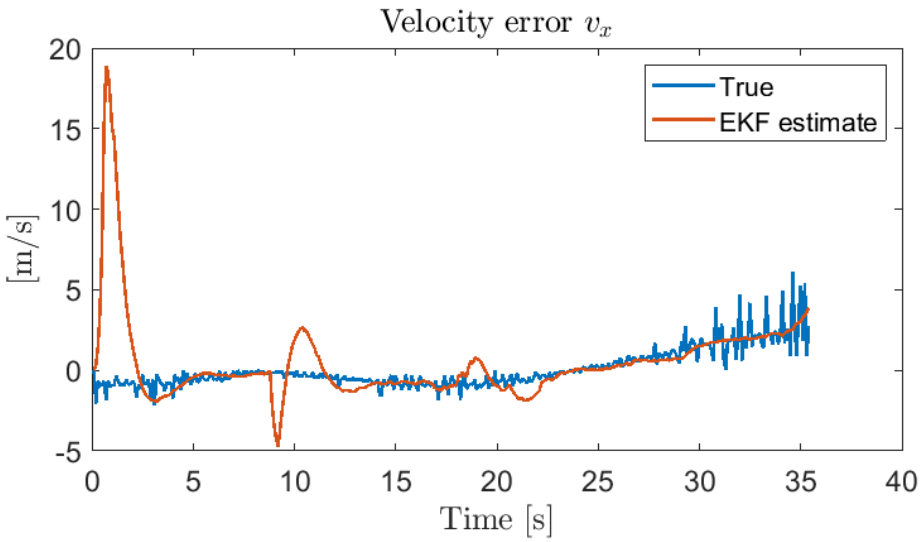
# AOA: position error vs iterations

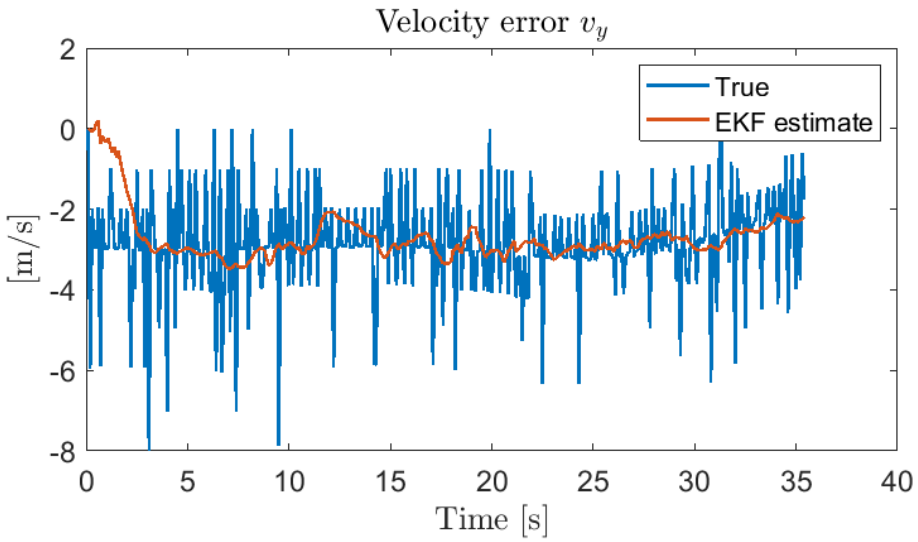


# TDOA + AOA: 1 iteration

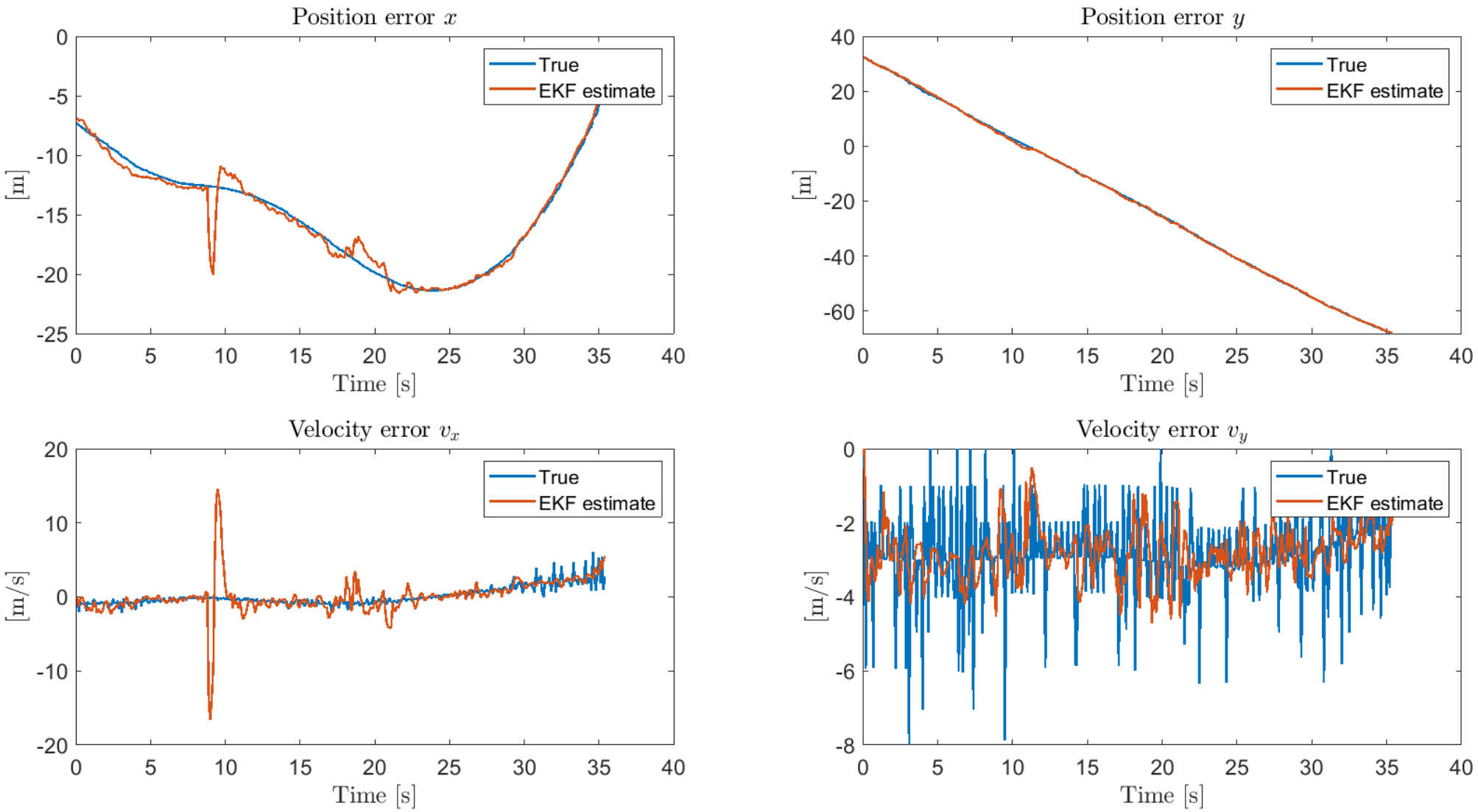


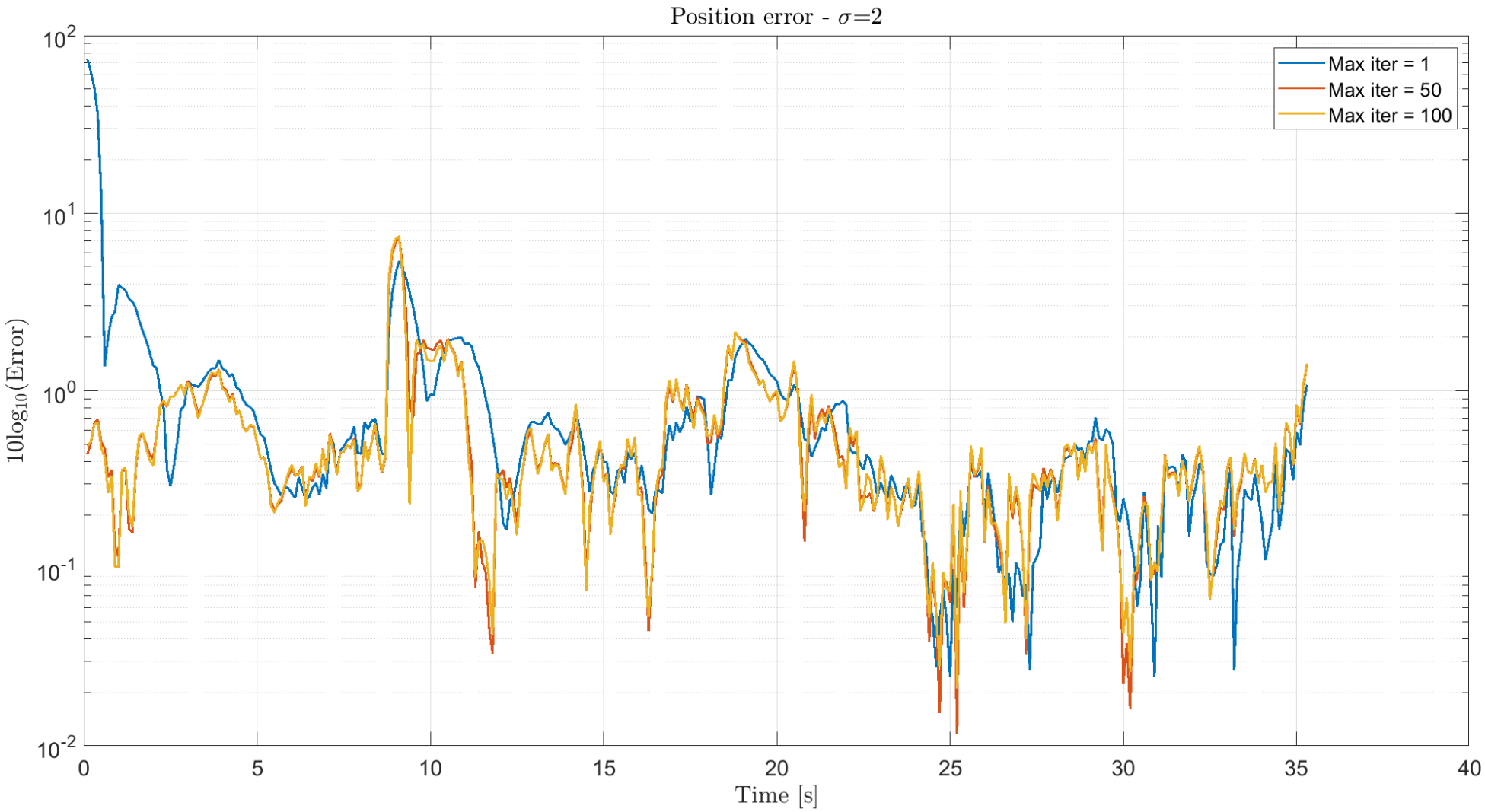



# TDOA + AOA: 100 NLS iterations



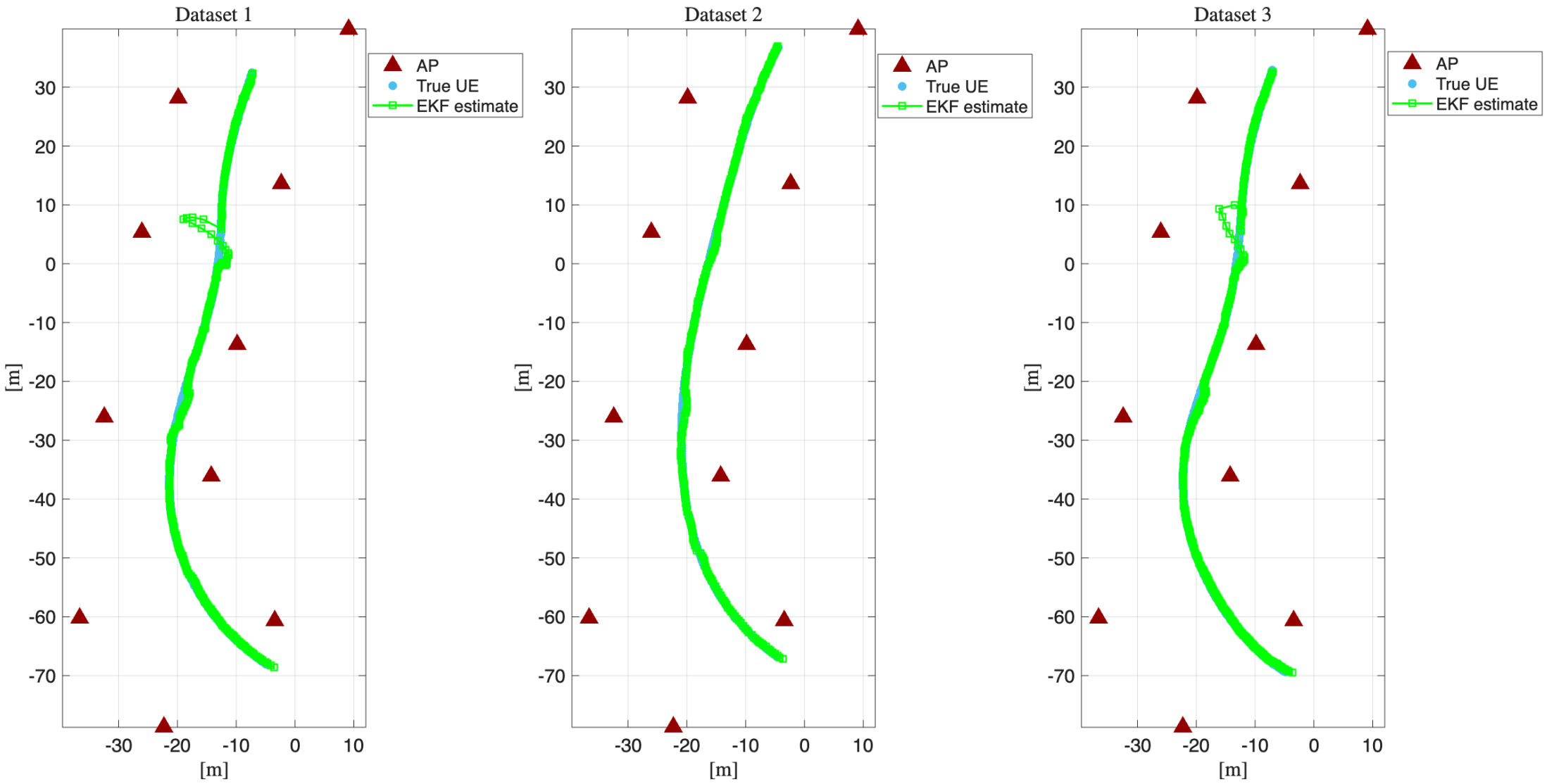
# TDOA + AOA: position error vs iterations



# Conclusions

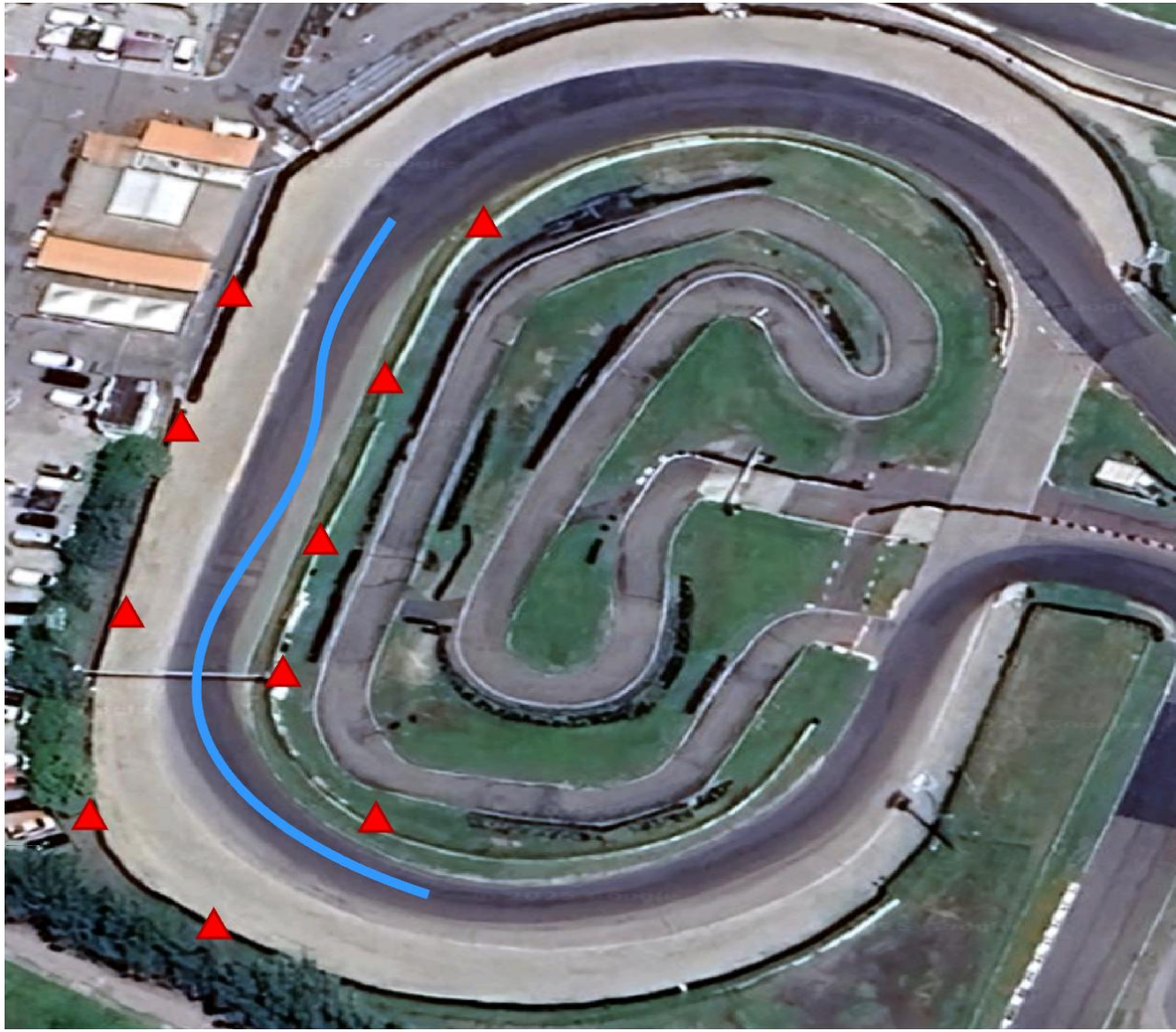
04

# Datasets comparison



# Conclusions:

- AOA and TDOA methods provide reliable localization within the NCV model limits and only after initial lock-on on all three datasets;
- If the covariance matrix is not tuned properly, the EKF may require several iterations before converging to the vehicle position;
- High accuracy can be achieved through iterative refinement with the Gauss–Newton method;
- The combination of the measurement of AOA and TDOA yields the best performance;
- For real-time applications, the number of Gauss–Newton iterations can be kept low, balancing accuracy and computational efficiency.



*Trajectory example over Castelletto di Branduzzo race track*

# Thank you for your attention!